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MOBILE ROBOT

Abstract

A mobile robot includes a body, a pair of front drive wheels at a front portion of the body, a pair of rear drive wheels at a rear portion of the body, four auxiliary wheels on a lower surface of the body, and a wheel switching device to switch the front drive wheels and the rear drive wheels into four-wheel mode or two-wheel mode. In four-wheel mode, the front drive wheels and the rear drive wheels contact a moving surface, and the four auxiliary wheels are spaced apart from the moving surface. In two-wheel mode, the front drive wheels are spaced apart from the moving surface and the rear drive wheels move to a center of the body and the body is lowered so that the rear drive wheels and the four auxiliary wheels contact the moving surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under § 365(c), of International Application No.

PCT/KR2025/002708, filed on Feb. 26, 2025, which is based on and claims priority to Korean Patent Application No. 10-2024-0029769, filed on Feb. 29, 2024, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to a mobile robot, and more particularly, to a mobile robot capable of traveling by selectively switching between a two-wheel mode and a four-wheel mode.

2. Description of Related Art

[0003] With the advancement of robot technology, mobile robots are increasingly used in various fields such as manufacturing, logistics, service, and the like.

[0004] Mobile robots may include outdoor mobile robots for outdoor traveling and indoor mobile robots for indoor traveling.

[0005] Outdoors, the moving space is relatively wide and there are many uneven sections, so outdoor mobile robots often include a large support polygon and four drive wheels. Therefore, the outdoor mobile robots may travel stably outdoors by utilizing the driving force of the four drive wheels.

[0006] Indoors, the moving space is narrow and complex, so indoor mobile robots often include a small support polygon and two drive wheels that can rotate in place. Therefore, the indoor mobile robots may travel indoors where the space is narrow and complex.

SUMMARY

[0007] According to an aspect of the disclosure, a mobile robot includes: a body; a pair of front drive wheels at a front portion of the body; a pair of rear drive wheels at a rear portion of the body; four auxiliary wheels on a lower surface of the body; a front wheel switching device including a front linear actuator disposed at an upper portion of the body; and a rear wheel switching device including a rear linear actuator disposed at a lower portion of the body, wherein the front wheel switching device and the rear wheel switching device are configured to switch the pair of front drive wheels and the pair of rear drive wheels between a four-wheel mode and a two-wheel mode, wherein, in the four-wheel mode, the pair of front drive wheels and the pair of rear drive wheels are in contact with a moving surface, and the four auxiliary wheels are spaced apart from the moving surface, and wherein, in the two-wheel mode, the pair of front drive wheels are spaced apart from the moving surface, the pair of rear drive wheels are in a center of the body, and the pair of rear drive wheels and the four auxiliary wheels are in contact with the moving surface.

[0008] The front wheel switching device may be configured to move the pair of front drive wheels up and down with respect to the body; and the rear wheel switching device is configured to move the pair of rear drive wheels forward and backward with respect to the body.

[0009] The front linear actuator may be configured to move the pair of front drive wheels up and

down.

[0010] The front wheel switching device further includes: an upper arm including a first end connected to the front linear actuator and a second end connected to the pair of front drive wheels; and a lower arm including a first end connected to the lower portion of the body and a second end connected to the pair of front drive wheels.

[0011] The front linear actuator includes: a front motor; a front screw configured to be rotated by the front motor; and a front nut that is screw-coupled to the front screw and to which the first end of the upper arm is connected, and rotation of the front motor causes the first end of the upper arm to move linearly along the front screw.

[0012] The upper arm may include a suspension.

[0013] The suspension includes a coil spring and a shock absorber.

[0014] The rear linear actuator may be configured to move the pair of rear drive wheels forward and backward with respect to the body.

[0015] The body may include: an upper support plate; a lower support plate below the upper support plate; and an inclined block on an upper surface of the lower support plate, the inclined block including an inclined surface, and the rear linear actuator is on the inclined surface of the inclined block.

[0016] The rear wheel switching device may further include a support bar including a first end connected to the rear linear actuator and a second end connected to the pair of rear drive wheels.

[0017] The rear linear actuator may include: a rear motor; a rear screw configured to be rotated by the rear motor; and a rear nut which is screw-coupled to the rear screw and is connected to the support bar, and rotation of the rear motor causes the support bar to move linearly along the rear screw.

[0018] The rear wheel switching device may further include: a fixed plate on the rear linear actuator; a movable plate parallel to the fixed plate and spaced apart from the fixed plate by a certain distance, wherein the pair of rear drive wheels are on the moveable plate; an upper link including a first end rotatably attached to the fixed plate and a second end rotatably attached to the movable plate; a lower link below the upper link, the lower link including a first end rotatably attached to the fixed plate and a second end rotatably attached to the movable plate; and a coil spring between the upper link and the lower link.

[0019] The rear wheel switching device may include: a fixed plate on the rear linear actuator; a movable plate parallel to the fixed plate and spaced apart from the fixed plate by a certain distance, wherein one of the pair of rear drive wheels is on the moveable plate; and a suspension between the movable plate and the fixed plate.

[0020] Each of the pair of front drive wheels and each of the pair of rear drive wheels may include an in-wheel motor.

[0021] Each of the four auxiliary wheels may include an omni wheel.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0022] The above and other aspects and features of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0023] FIG. 1 is a side view illustrating a mobile robot according to one or more embodiments of the disclosure;

[0024] FIG. 2 is a plan view illustrating the mobile robot of FIG. 1 according one or more embodiments of the disclosure;

[0025] FIG. 3 is a side view illustrating a state in which the mobile robot of FIG. 1 is switched to a

two-wheel mode according to one or more embodiments of the disclosure;

[0026] FIG. 4 is a perspective view illustrating a mobile robot according to one or more embodiments of the disclosure;

[0027] FIG. 5 is a side view illustrating the mobile robot of FIG. 4 according to one or more embodiments of the disclosure;

[0028] FIG. 6 is a front view illustrating the mobile robot of FIG. 4 according to one or more embodiments of the disclosure;

[0029] FIG. 7 is a rear view illustrating the mobile robot of FIG. 4 according to one or more embodiments of the disclosure;

[0030] FIG. 8 is a plan view illustrating the mobile robot of FIG. 4 according to one or more embodiments of the disclosure without an upper support plate;

[0031] FIG. 9 is a perspective view illustrating a front linear actuator of a mobile robot according to one or more embodiments of the disclosure;

[0032] FIG. 10 is a perspective view illustrating a rear linear actuator of a mobile robot according to one or more embodiments of the disclosure;

[0033] FIG. 11A is a partial view illustrating a left support bar of a rear wheel switching device of a mobile robot according to one or more embodiments of the disclosure;

[0034] FIG. 11B is a partial view illustrating a left support bar of a rear wheel switching device of a mobile robot according to one or more embodiments of the disclosure when a rear drive wheel is raised;

[0035] FIG. 11C is a partial view illustrating a left support bar of a rear wheel switching device of a mobile robot according to one or more embodiments of the disclosure when a rear drive wheel is lowered;

[0036] FIG. 12 is a partial view illustrating a left support bar of a rear wheel switching device of a mobile robot according to one or more embodiments of the disclosure;

[0037] FIG. 13 is a perspective view illustrating an auxiliary wheel of a mobile robot according to one or more embodiments of the disclosure;

[0038] FIG. 14 is a side view illustrating the auxiliary wheel of FIG. 13 according to one or more embodiments of the disclosure;

[0039] FIG. 15 is a perspective view illustrating a mobile robot according to one or more embodiments of the disclosure, which has been switched to a two-wheel mode;

[0040] FIG. 16 is a side view illustrating the mobile robot of FIG. 15 according to one or more embodiments of the disclosure;

[0041] FIG. 17 is a front view illustrating the mobile robot of FIG. 15 according to one or more embodiments of the disclosure; and

[0042] FIG. 18 is a block diagram illustrating a mobile robot according to one or more embodiments of the disclosure.

DETAILED DESCRIPTION

[0043] The various embodiments described herein and terms used herein are not intended to limit the technical features described in this disclosure to specific embodiments, but should be understood to include various modifications, equivalents, or alternatives of the embodiments.

[0044] In connection with the description of the drawings, similar reference numbers may be used for similar or related components.

[0045] The singular form of a noun corresponding to an item may include one or more of the above item, unless the relevant context clearly indicates otherwise.

[0046] As used herein, each of phrases such as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” “at least one of A, B, C” may include any one of the items listed together with the corresponding phrase, or any possible combination thereof.

[0047] The term “and/or” includes any element of a plurality of related described elements or a combination of a plurality of related described elements.

[0048] Terms such as “first,” “second,” “primary,” or “secondary” may be used simply to distinguish one component from other components, and do not limit the corresponding components in other respects (e.g., importance or order).

[0049] When it is mentioned that one (e.g., first) component is “coupled” or “connected” to another (e.g., second) component with or without terms “functionally” or “communicatively”, it means that the one component can be connected to the another component directly (e.g., wired), wirelessly, or through a third component.

[0050] Terms such as “include” or “have” are intended to designate the presence of features, numbers, steps, operations, components, parts, or combinations thereof described in the embodiment, but do not preclude the presence or addition of one or more other features, numbers, steps, operations, components, parts, or combination thereof.

[0051] When a component is said to be “connected,” “coupled,” “supported,” or “in contact” with another component, this means not only cases where the components are directly connected, coupled, supported, or contacted, but also cases where the components are indirectly connected, coupled, supported, or contacted through a third component.

[0052] When a component is said to be located “on” other component, this includes not only cases where the component is in contact with the other component, but also cases where another component exists between the two components.

[0053] Further, the terms “leading end”, “rear end”, “upper side”, “lower side”, “top end”, “bottom end”, etc. used in the disclosure are defined with reference to the drawings. However, the shape and position of each component are not limited by the terms.

[0054] The disclosure relates to a mobile robot having high traveling performance both indoors and outdoors.

[0055] Hereinafter, a mobile robot **1** according to one or more embodiments of the disclosure will be described in detail with reference to FIGS. **1**, **2**, and **3**.

[0056] FIG. **1** is a side view illustrating a mobile robot **1** according to one or more embodiments of the disclosure. FIG. **2** is a plan view illustrating the mobile robot **1** of FIG. **1** according one or more embodiments of the disclosure. FIG. **3** is a side view illustrating a state in which the mobile robot **1** of FIG. **1** is switched to a two-wheel mode according to one or more embodiments of the disclosure. For reference, in FIG. **2**, an upper support plate **11** of the mobile robot **1** is removed.

[0057] Referring to FIGS. **1**, **2**, and **3**, a mobile robot **1** according to one or more embodiments of the disclosure may include a body **10**, a pair of front drive wheels **20**, a pair of rear drive wheels **30**, four auxiliary wheels **40**, and a wheel switching device **50** and **60**.

[0058] The body **10** may be configured to support and fix the pair of front drive wheels **20**, the pair of rear drive wheels **30**, the four auxiliary wheels **40**, and the wheel switching device **50** and **60**.

[0059] A module **3** corresponding to the purpose of the mobile robot **1** may be disposed on the upper surface of the body **10**. For example, when the mobile robot **1** is used for delivery, a delivery module may be disposed on the upper surface of the body **10**. When the mobile robot **1** is used for guidance, a guidance module may be disposed on the upper surface of the body **10**.

[0060] The body **10** may include an upper support plate **11** and a lower support plate **12**.

[0061] The upper support plate **11** and the lower support plate **12** may be spaced apart from each other by a certain distance in the vertical direction. A plurality of support columns **13** may be disposed between the upper support plate **11** and the lower support plate **12**. For example, the upper support plate **11** and the lower support plate **12** may be connected by the plurality of support columns **13**.

[0062] The upper support plate **11** may be formed as a flat plate having an approximately rectangular shape. The lower support plate **12** may be formed in a shape corresponding to the upper support plate **11**. The lower support plate **12** may be formed as a flat plate having an approximately rectangular shape.

[0063] The pair of front drive wheels **20** may be disposed at the front portion of the body **10**. Each

of the pair of front drive wheels **20** may include a front in-wheel motor **21**. The front in-wheel motor **21** may be disposed at the center of the front drive wheel **20**. The front drive wheel **20** may be rotated by the front in-wheel motor **21**.

[0064] The pair of front drive wheels **20** may be disposed on the body **10** to selectively contact a moving surface **100**. When the mobile robot **1** is in a two-wheel mode, the pair of front drive wheels **20** do not contact the moving surface **100**. When the mobile robot **1** is in a four-wheel mode, the pair of front drive wheels **20** contact the moving surface **100**. Here, the moving surface **100** refers to a surface such as a road, a sidewalk, a floor of a building, a hallway, a floor of an elevator, and the like on which the mobile robot **1** may move.

[0065] When the pair of front drive wheels **20** contact the moving surface **100**, the pair of front drive wheels **20** may protrude below the lower surface of the body **10**.

[0066] When the pair of front drive wheels **20** contact the moving surface **100**, the pair of front drive wheels **20** may protrude lower than the four auxiliary wheels **40** disposed on the lower surface of the body **10**. Therefore, when the pair of front drive wheels **20** contact the moving surface **100**, the four auxiliary wheels **40** do not contact the moving surface **100**.

[0067] The pair of rear drive wheels **30** may be disposed on the rear portion of the body **10**. Each of the pair of rear drive wheels **30** may include a rear in-wheel motor **31**. The rear in-wheel motor **31** may be disposed at the center of the rear drive wheel **30**. The rear drive wheel **30** may be rotated by the rear in-wheel motor **31**.

[0068] The pair of rear drive wheels **30** may be disposed on the body **10** so as to contact the moving surface **100**. When the mobile robot **1** is in the two-wheel mode and the four-wheel mode, the pair of rear drive wheels **30** contact the moving surface **100**.

[0069] The pair of rear drive wheels **30** may be disposed so as to protrude lower than the lower surface of the body **10**.

[0070] In the case of the four-wheel mode in which the pair of front drive wheels **20** contact the moving surface **100**, the pair of rear drive wheels **30** may protrude lower than the four auxiliary wheels **40** disposed on the lower surface of the body **10**. Therefore, when the pair of front drive wheels **20** contact the moving surface **100**, the four auxiliary wheels **40** do not contact the moving surface **100**.

[0071] In the case of the two-wheel mode in which the pair of front drive wheels **20** do not contact the moving surface **100**, the pair of rear drive wheels **30** may protrude to the same height as the four auxiliary wheels **40** disposed on the lower surface of the body **10**. Accordingly, when the pair of front drive wheels **20** do not contact the moving surface **100**, the four auxiliary wheels **40** may contact the moving surface **100** together with the pair of rear drive wheels **30**.

[0072] The four auxiliary wheels **40** may be disposed on the lower surface of the body **10**. A wheel that can rotate in all directions may be used as the four auxiliary wheels **40**. For example, casters, omni wheels, mecanum wheels, etc. may be used as the auxiliary wheels **40**.

[0073] The wheel switching device **50** and **60** may be configured so that the mobile robot **1** may be switched into one of the four-wheel mode and the two-wheel mode. The wheel switching device **50** and **60** may be configured so that the positions of the pair of front drive wheels **20** and the pair of rear drive wheels **30** are switched into one of the four-wheel mode and the two-wheel mode.

[0074] The four-wheel mode is a state in which the pair of front drive wheels **20** and the pair of rear drive wheels **30** are in contact with the moving surface **100**. In the four-wheel mode, the four auxiliary wheels **40** are spaced apart from the moving surface **100**.

[0075] The two-wheel mode is a state in which the pair of front drive wheels **20** are spaced apart from the moving surface **100** and the pair of rear drive wheels **30** are in contact with the moving surface **100**. In the two-wheel mode, the four auxiliary wheels **40** are in contact with the moving surface **100**.

[0076] The wheel switching device may include a front wheel switching device **50** and a rear wheel switching device **60**.

[0077] The front wheel switching device **50** may be configured to move the pair of front drive wheels **20**. The front wheel switching device **50** may be configured to move the pair of front drive wheels **20** up and down with respect to the body **10**. In the four-wheel mode, the front wheel switching device **50** may lower the pair of front drive wheels **20** so that the pair of front drive wheels **20** come into contact with the moving surface **100**. In the two-wheel mode, the front wheel switching device **50** may raise the pair of front drive wheels **20** so that the pair of front drive wheels **20** are spaced apart from the moving surface **100**.

[0078] The front wheel switching device **50** may include a front linear actuator **51**. The front linear actuator **51** may be configured to raise and lower the pair of front drive wheels **20**. The pair of front drive wheels **20** may be raised above the lower surface of the body **10** or may be lowered below the lower surface of the body **10** by the front linear actuator **51**.

[0079] The front linear actuator **51** may be disposed on the upper portion of the body **10**. For example, the front linear actuator **51** may be disposed on the lower surface of the upper support plate **11** of the body **10**.

[0080] The front linear actuator **51** may include a front motor **52**, a front screw **53**, and a front nut **54**.

[0081] The front motor **52** may be configured to rotate the front screw **53**. The front motor **52** may be configured to rotate the front screw **53** in both directions. The front motor **52** may include a motor shaft **521**.

[0082] The front screw **53** may be disposed to rotate by the front motor **52**. The front screw **53** may be supported at both ends on the lower surface of the upper support plate **11** of the body **10**. The front screw **53** may be supported by a pair of support blocks **531** disposed on the lower surface of the upper support plate **11**.

[0083] A first end of the front screw **53** may be coupled to the motor shaft **521** of the front motor **52** by a coupling **522**. Therefore, when the motor shaft **521** rotates, the front screw **53** may rotate integrally with the motor shaft **521**.

[0084] The front nut **54** may be screw-coupled to the front screw **53**. Therefore, when the front screw **53** rotates, the front nut **54** may move linearly along the front screw **53**. For example, the front screw **53** and the front nut **54** may convert the rotation of the front motor **52** into the linear movement of the front nut **54**. The front screw **53** and the front nut **54** may include a lead screw and a nut or a ball screw and a ball nut.

[0085] The front wheel switching device **50** may include an upper arm **56** and a lower arm **57**.

[0086] The upper arm **56** may connect the front linear actuator **51** and the front drive wheels **20**. A first end of the upper arm **56** may be connected to the front linear actuator **51**, and a second end thereof may be connected to the front drive wheels **20**. The first end of the upper arm **56** may be rotatably connected to the front nut **54** of the front linear actuator **51**, and the second end of the upper arm **56** may be rotatably connected to the front drive wheels **20**.

[0087] The lower arm **57** may connect the body **10** and the front drive wheels **20**. A first end of the lower arm **57** may be connected to the lower portion of the body **10**, that is, the lower support plate **12** of the body **10**, and a second end thereof may be connected to the front drive wheels **20**. The first end of the lower arm **57** may be rotatably connected to the lower portion of the body **10**, and the second end of the lower arm **57** may be rotatably connected to the front drive wheels **20**.

[0088] When the front drive wheels **20** includes the front in-wheel motors **21**, the second end of the upper arm **56** and the second end of the lower arm **57** may be rotatably connected to a support bracket that supports the front in-wheel motors **21**.

[0089] The front wheel switching device **50** may include a pair of upper arms **56** and a pair of lower arms **57** corresponding to the pair of front drive wheels **20**. For example, the front wheel switching device **50** may include a left upper arm **56** and a left lower arm **57** that are connected to the left front drive wheel **20**, and a right upper arm **56** and a right lower arm **57** that are connected to the right front drive wheel **20**.

[0090] The pair of upper arms **56** and the pair of lower arms **57** may be configured so that the pair of front drive wheels **20** protrude to the maximum extent toward the front of the body **10**. When the pair of front drive wheels **20** are configured so that they protrude to the maximum extent toward the front of the body **10**, the area of the support polygon of the mobile robot **1** may be increased.

[0091] Therefore, when the front motor **52** rotates, the first end of the upper arm **56** may move linearly along the front screw **53**. For example, when the front motor **52** rotates, the front screw **53** may rotate. When the front screw **53** rotates, the front nut **54** screw-coupled to the front screw **53** may move linearly along the front screw **53**. Because the first end of the upper arm **56** is rotatably connected to the front nut **54**, when the front screw **53** rotates, the first end of the upper arm **56** may move linearly along the front screw **53**.

[0092] When the upper arm **56** moves linearly along the front screw **53**, the front drive wheels **20** may be rotated at a certain angle relative to the body **10** by the upper arm **56** and the lower arm **57**. When the front motor **52** rotates in one direction, the front drive wheels **20** may swing upward, that is, counter-clockwise, relative to the body **10**, and rise. When the front motor **52** rotates in the opposite direction, the front drive wheels **20** may swing downward, that is, clockwise, relative to the body **10**, and descend.

[0093] The rear wheel switching device **60** may be configured to move the pair of rear drive wheels **30**. The rear wheel switching device **60** may be configured to move the pair of rear drive wheels **30** back and forth relative to the body **10**.

[0094] In the four-wheel mode, the rear wheel switching device **60** may move the pair of rear drive wheels **30** toward the rear of the body **10** so that the pair of rear drive wheels **30** protrude from the rear end of the body **10**. The pair of rear drive wheels **30** may be disposed to protrude to the maximum extent toward the rear of the body **10**. When the pair of rear drive wheels **30** are configured so that they protrude to the maximum extent from the rear end of the body **10**, the area of the support polygon of the mobile robot **1** may be increased.

[0095] In the two-wheel mode, the rear wheel switching device **60** may move the pair of rear drive wheels **30** to the center of the body **10** so that the pair of rear drive wheels **30** are positioned at the center of the body **10**, and the body **10** is lowered. Then, the four auxiliary wheels **40** disposed on the lower surface of the body **10** come contact with the moving surface **100**. At this time, the pair of rear drive wheels **30** may be positioned at the middle of the four auxiliary wheels **40**. For example, as illustrated in FIG. 3, two auxiliary wheels **40** may be positioned on the left side of the pair of rear drive wheels **30**, and the other two auxiliary wheels **40** may be positioned on the right side of the pair of rear drive wheels **30**.

[0096] The rear wheel switching device **60** may include a rear linear actuator **61**. The rear linear actuator **61** may be configured to move the pair of rear drive wheels **30** back and forth with respect to the body **10**. The pair of rear drive wheels **30** may be positioned at the rear of the body **10** or at the center of the body **10** by the rear linear actuator **61**.

[0097] The rear linear actuator **61** may be disposed at the lower portion of the body **10**. For example, the rear linear actuator **61** may be disposed on the upper surface of the lower support plate **12** of the body **10**.

[0098] The body **10** may include an inclined block **15** disposed on the upper surface of the lower support plate **12**. The rear linear actuator **61** may be disposed on an inclined surface **151** of the inclined block **15**.

[0099] The inclined block **15** may be formed to have the inclined surface **151** that is inclined upwardly with respect to the lower support plate **12**. The inclined block **15** may be disposed so that the high point of the inclined surface **151** is adjacent to the front end of the body **10** and the low point of the inclined surface **151** is adjacent to the rear end of the body **10**. The angle of the inclined surface **151** may be defined so that when the pair of rear drive wheels **30** are positioned at the center of the body **10**, the four auxiliary wheels **40** disposed on the lower surface of the body **10** come into contact with the moving surface **100**.

[0100] The rear linear actuator **61** may include a rear motor **62**, a rear screw **63**, and a rear nut **64**.
[0101] The rear motor **62** may be configured to rotate the rear screw **63**. The rear motor **62** may be configured to rotate the rear screw **63** in both directions. The rear motor **62** may include a motor shaft **621**.

[0102] The rear screw **63** may be disposed to be rotated by the rear motor **62**. The rear screw **63** may be disposed so that both ends of the rear screw **63** are supported on the inclined surface **151** of the inclined block **15** on the lower support plate **12**. The rear screw **63** may be supported by a pair of support blocks **631** disposed on the inclined surface **151** of the inclined block **15**.

[0103] A first end of the rear screw **63** may be coupled to the motor shaft **621** of the rear motor **62** by a coupling **622**. Therefore, when the motor shaft **621** rotates, the rear screw **63** may rotate integrally with the motor shaft **621**.

[0104] The rear nut **64** may be screw-coupled to the rear screw **63**. Therefore, when the rear screw **63** rotates, the rear nut **64** may move linearly along the rear screw **63**. For example, the rear screw **63** and the rear nut **64** may be configured to convert the rotational motion of the rear motor **62** into the linear motion of the rear nut **64**. The rear screw **63** and the rear nut **64** may include a lead screw and a nut or a ball screw and a ball nut.

[0105] The rear wheel switching device **60** may include a support bar **66** configured to connect the pair of rear drive wheels **30** and the rear linear actuator **61**. A first end of the support bar **66** may be connected to the rear linear actuator **61**, and a second end thereof may be connected to the pair of rear drive wheels **30**.

[0106] The support bar **66** may be connected to the rear nut **64** of the rear linear actuator **61**. For example, the rear nut **64** may be disposed at the center of the support bar **66**. Therefore, when the rear nut **64** moves linearly by the rear screw **63**, the support bar **66** may move linearly along the rear screw **63** together with the rear nut **64**.

[0107] The pair of rear drive wheels **30** may be disposed at both ends of the support bar **66**. When the pair of rear drive wheels **30** include the rear in-wheel motors **31**, both ends of the support bar **66** may be connected to support brackets configured to support the rear in-wheel motors **31**, respectively.

[0108] Therefore, when the rear motor **62** rotates, the support bar **66** may move linearly along the rear screw **63**. For example, when the rear motor **62** rotates, the rear screw **63** may rotate. When the rear screw **63** rotates, the rear nut **64** screw-coupled to the rear screw **63** may move linearly along the rear screw **63**. Because the support bar **66** is connected to the rear nut **64**, when the rear screw **63** rotates, the support bar **66** may move linearly along the rear screw **63**.

[0109] When the support bar **66** moves linearly along the rear screw **63** disposed on the inclined block **15**, the body **10** may be raised or lowered with respect to the moving surface **100** by the inclined block **15**. When the rear motor **62** rotates in one direction, the rear drive wheels **30** may move to the center of the body **10** and the body **10** may be lowered. When the rear motor **62** rotates in the opposite direction, the rear drive wheels **30** may move rearward and the body **10** may be raised.

[0110] In the four-wheel mode illustrated in FIGS. **1** and **2**, the pair of front drive wheels **20** and the pair of rear drive wheels **30** come into contact with the moving surface **100**. In this case, the pair of front drive wheels **20** may protrude to the maximum from the front end of the body **10**, and the pair of rear drive wheels **30** may protrude to the maximum from the rear end of the body **10**. Accordingly, the area of the support polygon of the mobile robot **1** may be increased so that the mobile robot **1** may stably move on an uneven moving surface **100**, such as outdoors.

[0111] In the two-wheel mode illustrated in FIG. **3**, the pair of front drive wheels **20** are spaced apart from the moving surface **100**, and the pair of rear drive wheels **30** and the four auxiliary wheels **40** come into contact with the moving surface **100**. In this case, the pair of front drive wheels **20** are raised and positioned inside the body **10**, and the pair of rear drive wheels **30** are positioned at the center of the body **10**. Therefore, the area of the support polygon of the mobile

robot **1** may be minimized, so that the mobile robot **1** may move efficiently in a narrow space such as indoors. In the two-wheel mode, the mobile robot **1** may rotate in place by the pair of rear drive wheels **30**.

[0112] FIG. **4** is a perspective view illustrating a mobile robot **1** according to one or more embodiments of the disclosure. FIG. **5** is a side view illustrating the mobile robot **1** of FIG. **4** according to one or more embodiments of the disclosure. FIG. **6** is a front view illustrating the mobile robot **1** of FIG. **4** according to one or more embodiments of the disclosure. FIG. **7** is a rear view illustrating the mobile robot **1** of FIG. **4** according to one or more embodiments of the disclosure. FIG. **8** is a plan view illustrating the mobile robot **1** of FIG. **4** according to one or more embodiments of the disclosure without an upper support plate **11**. FIG. **9** is a perspective view illustrating a front linear actuator **51** of a mobile robot **1** according to one or more embodiments of the disclosure. FIG. **10** is a perspective view illustrating a rear linear actuator **61** of a mobile robot **1** according to one or more embodiments of the disclosure.

[0113] Referring to FIGS. **4** to **8**, a mobile robot **1** may include a body **10**, a pair of front drive wheels **20**, a pair of rear drive wheels **30**, four auxiliary wheels **40**, and wheel switching devices **50** and **60**.

[0114] The body **10** may be configured to support and fix the pair of front drive wheels **20**, the pair of rear drive wheels **30**, the four auxiliary wheels **40**, and the wheel switching devices **50** and **60**.

[0115] A module **3** corresponding to the purpose of the mobile robot **1** may be disposed on the upper surface of the body **10** (see FIG. **1**). For example, a delivery module, a guidance module, etc. may be disposed on the upper surface of the body **10**.

[0116] The body **10** may include an upper support plate **11** and a lower support plate **12**.

[0117] The upper support plate **11** and the lower support plate **12** may be spaced apart from each other by a certain distance in the vertical direction. A plurality of support columns **13** may be disposed between the upper support plate **11** and the lower support plate **12**. The upper support plate **11** and the lower support plate **12** may be connected by the plurality of support columns **13**. For example, six support columns **13** may be disposed between the upper support plate **11** and the lower support plate **12**.

[0118] The upper support plate **11** may be formed as a flat plate having an approximately rectangular shape. The lower support plate **12** may be formed in a shape corresponding to the upper support plate **11**. The lower support plate **12** may be formed as a flat plate having an approximately rectangular shape. The lower support plate **12** may be formed smaller than the upper support plate **11**.

[0119] The pair of front drive wheels **20** may be disposed at the front portion of the body **10**. The pair of front drive wheels **20** may be disposed so as to protrude forward from the front surface of the body **10**. Each of the pair of front drive wheels **20** may include a front in-wheel motor **21**. The front in-wheel motor **21** may be disposed at the center of the front drive wheel **20**. The front drive wheel **20** may be rotated by the front in-wheel motor **21**.

[0120] The pair of front drive wheels **20** may be disposed in the body **10** to selectively contact a moving surface **100**. When the mobile robot **1** is in the two-wheel mode, the pair of front drive wheels **20** may not contact the moving surface **100**. When the mobile robot **1** is in the four-wheel mode, the pair of front drive wheels **20** may contact the moving surface **100**. Here, the moving surface **100** refers to a surface such as a road, a sidewalk, a floor of a building, a hallway, a floor of an elevator, and the like on which the mobile robot **1** may move.

[0121] When the pair of front drive wheels **20** come into contact with the moving surface **100**, the pair of front drive wheels **20** may protrude downward from the lower surface of the body **10** and forward from the front surface of the body **10**.

[0122] When the pair of front drive wheels **20** come into contact with the moving surface **100**, the pair of front drive wheels **20** may protrude lower than the four auxiliary wheels **40** disposed on the lower surface of the body **10**. Therefore, when the pair of front drive wheels **20** come into contact

with the moving surface **100**, the four auxiliary wheels **40** may not contact the moving surface **100**.

[0123] When the mobile robot **1** is in the two-wheel mode, the pair of front drive wheels **20** may be positioned inside the body **10** and may not protrude forward of the body **10**.

[0124] The pair of rear drive wheels **30** may be disposed at the rear portion of the body **10**. Each of the pair of rear drive wheels **30** may include a rear in-wheel motor **31**. The rear in-wheel motor **31** may be disposed at the center of the rear drive wheel **30**. The rear drive wheel **30** may be rotated by the rear in-wheel motor **31**.

[0125] The pair of rear drive wheels **30** may be disposed in the body **10** so as to contact the moving surface **100**. When the mobile robot **1** travels, regardless of the two-wheel mode or the four-wheel mode, the pair of rear drive wheels **30** may always contact the moving surface **100**.

[0126] The pair of rear drive wheels **30** may be disposed so as to protrude downward from the lower surface of the body **10** and backward from the rear surface of the body **10**.

[0127] In the case of the four-wheel mode in which the pair of front drive wheels **20** contact the moving surface **100**, the pair of rear drive wheels **30** may protrude lower than the four auxiliary wheels **40** disposed on the lower surface of the body **10**. Therefore, when the pair of front drive wheels **20** contact the moving surface **100**, the four auxiliary wheels **40** may not contact the moving surface **100**.

[0128] In the case of the two-wheel mode in which the pair of front drive wheels **20** do not contact the moving surface **100**, the pair of rear drive wheels **30** may protrude to the same height as the four auxiliary wheels **40** disposed on the lower surface of the body **10**. Accordingly, when the pair of front drive wheels **20** do not contact the moving surface **100**, the four auxiliary wheels **40** may contact the moving surface **100** together with the pair of rear drive wheels **30**.

[0129] The wheel switching devices **50** and **60** may be configured so that the mobile robot **1** may be switched into one of the four-wheel mode and the two-wheel mode. The wheel switching devices **50** and **60** may be configured so that the positions of the pair of front drive wheels **20** and the pair of rear drive wheels **30** are switched into one of the four-wheel mode and the two-wheel mode.

[0130] The four-wheel mode is a state in which the pair of front drive wheels **20** and the pair of rear drive wheels **30** are in contact with the moving surface **100**. The pair of front drive wheels **20** may protrude forward from the front surface of the body **10**, and the pair of rear drive wheels **30** may protrude backward from the rear surface of the body **10**. In the four-wheel mode, the four auxiliary wheels **40** may be spaced apart from the moving surface **100**.

[0131] The two-wheel mode is a state in which the pair of front drive wheels **20** are spaced apart from the moving surface **100** and the pair of rear drive wheels **30** are in contact with the moving surface **100**. The pair of front drive wheels **20** may be located inside the body **10** and may not protrude from the front surface of the body **10**. The pair of rear drive wheels **30** may be located in the center of the body **10**. In the two-wheel mode, the four auxiliary wheels **40** may be in contact with the moving surface **100**. In the two-wheel mode, when the pair of rear drive wheels **30** move to the center of the body **10**, the body **10** may be lowered so that the four auxiliary wheels **40** come into contact with the moving surface **100**.

[0132] The wheel switching devices may include a front wheel switching device **50** and a rear wheel switching device **60**.

[0133] The front wheel switching device **50** may be configured to move the pair of front drive wheels **20**. The front wheel switching device **50** may be configured to move the pair of front drive wheels **20** up and down with respect to the body **10**.

[0134] In the four-wheel mode, the front wheel switching device **50** may lower the pair of front drive wheels **20** so that the pair of front drive wheels **20** come into contact with the moving surface **100**. The front wheel switching device **50** may lower the pair of front drive wheels **20** so that the pair of front drive wheels **20** protrude from the front surface and the lower surface of the body **10**.

[0135] In the two-wheel mode, the front wheel switching device **50** may raise the pair of front drive

wheels **20** so that the pair of front drive wheels **20** are spaced apart from the moving surface **100**. The front wheel switching device **50** may raise the pair of front drive wheels **20** so that the pair of front drive wheels **20** are positioned inside the body **10**, that is, in the space between the upper support plate **11** and the lower support plate **12**. In this configuration, the pair of front drive wheels **20** may not protrude from the front surface of the body **10**, that is, from the front ends of the upper support plate **11** and the lower support plate **12**.

[0136] The front wheel switching device **50** may include a front linear actuator **51**. The front linear actuator **51** may be configured to raise and lower the pair of front drive wheels **20**. The pair of front drive wheels **20** may be positioned above or protrude below the lower surface of the body **10** by the front linear actuator **51**.

[0137] The front linear actuator **51** may be disposed on the upper portion of the body **10**. For example, the front linear actuator **51** may be disposed on the lower surface of the upper support plate **11** of the body **10**.

[0138] Referring to FIGS. **5**, **8**, and **9**, the front linear actuator **51** may include a front motor **52**, a front screw **53**, and a front nut **54**.

[0139] The front motor **52** may be disposed on the lower surface of the upper support plate **11**. The front motor **52** may be configured to rotate the front screw **53**. The front motor **52** may be configured to rotate the front screw **53** in both directions. The front motor **52** may include a motor shaft **521**.

[0140] The front screw **53** may be disposed to be rotated by the front motor **52**. The front screw **53** may be supported at both ends on the lower surface of the upper support plate **11** of the body **10**. The front screw **53** may be supported by a pair of support blocks **531** disposed on the lower surface of the upper support plate **11**.

[0141] A first end of the front screw **53** may be coupled with the motor shaft **521** of the front motor **52** by a coupling **522**. Therefore, when the motor shaft **521** rotates, the front screw **53** may rotate integrally with the motor shaft **521**.

[0142] The front nut **54** may be screw-coupled to the front screw **53**. Therefore, when the front screw **53** rotates, the front nut **54** may move linearly along the front screw **53**. For example, the front screw **53** and the front nut **54** may convert the rotation of the front motor **52** into linear movement of the front nut **54**. The front screw **53** and the front nut **54** may be configured as a lead screw and a nut or a ball screw and a ball nut.

[0143] The front wheel switching device **50** may include an upper arm **56** and a lower arm **57**.

[0144] The upper arm **56** may connect the front linear actuator **51** and the front drive wheel **20**. the first end of the upper arm **56** may be connected to the front linear actuator **51**, and the second end thereof may be connected to the front drive wheel **20**. The first end of the upper arm **56** may be rotatably connected to the front nut **54** of the front linear actuator **51**, and the second end of the upper arm **56** may be rotatably connected to the front drive wheel **20**.

[0145] The upper arm **56** may be configured as a suspension. The suspension may be configured to absorb shock applied to the front drive wheel **20**. The suspension may include a coil spring and a shock absorber. The coil spring and the shock absorber may be arranged in a straight line.

[0146] The lower arm **57** may connect the body **10** and the front drive wheel **20**. A first end of the lower arm **57** may be connected to the lower portion of the body **10**, that is, to the lower support plate **12** of the body **10**, and the second end thereof may be connected to the front drive wheel **20**. The first end of the lower arm **57** may be rotatably connected to the lower portion of the body **10**, and the second end of the lower arm **57** may be rotatably connected to the front drive wheel **20**.

[0147] When the front drive wheels **20** include the front in-wheel motor **21**, the second end of the upper arm **56** and the second end of the lower arm **57** may be rotatably connected to a support bracket **23** that supports the front in-wheel motor **21**.

[0148] The front wheel switching device **50** may include a pair of upper arms **56** and a pair of lower arms **57** corresponding to the pair of front drive wheels **20**. For example, the front wheel

switching device **50** may include a left upper arm **56** and a left lower arm **57** that are connected to the left front drive wheel **20**, and a right upper arm **56** and a right lower arm **57** that are connected to the right front drive wheel **20**.

[0149] The front wheel switching device **50** may include a middle bar **55** connecting the pair of upper arms **56** and the front nut **54**. The front nut **54** may be connected to the center of the middle bar **55**, the first end of the left upper arm **56** may be connected to the left end of the middle bar **55**, and the first end of the right upper arm **56** may be connected to the right end of the middle bar **55**.

[0150] A left hinge **551** that rotatably supports the first end of the left upper arm **56** may be provided at the left end of the middle bar **55**. The left hinge **551** may be guided by a left LM guide **58**. The left LM guide **58** may be disposed on the lower surface of the upper support plate **11**.

[0151] A right hinge **552** that rotatably supports the first end of the right upper arm **56** may be provided at the right end of the middle bar **55**. The right hinge **552** may be guided by a right LM guide **59**. The right LM guide **59** may be disposed on the lower surface of the upper support plate **11**. The right LM guide **59** may be disposed parallel to the left LM guide **58**.

[0152] The second end of the left upper arm **56** may be rotatably disposed on a left support bracket **23** disposed at the front in-wheel motor **21** of the left front drive wheel **20**.

[0153] The second end of the right upper arm **56** may be rotatably disposed on a right support bracket **23** disposed at the front in-wheel motor **21** of the right front drive wheel **20**.

[0154] The first end of the left lower arm **57** may be rotatably supported by a left lower hinge **571** disposed on the lower support plate **12**.

[0155] The second end of the left lower arm **57** may be rotatably disposed on the left support bracket **23** disposed at the front in-wheel motor **21** of the left front drive wheel **20**.

[0156] The first end of the right lower arm **57** may be rotatably supported by a right lower hinge **571** disposed on the lower support plate **12**.

[0157] The second end of the right lower arm **57** may be rotatably disposed on the right support bracket **23** disposed at the front in-wheel motor **21** of the right front drive wheel **20**.

[0158] The front wheel switching device **50** may be configured as a MacPherson strut suspension.

[0159] The pair of upper arms **56** and the pair of lower arms **57** may be configured so that the pair of front drive wheels **20** protrude to the maximum extent toward the front of the body **10**. When the pair of front drive wheels **20** are configured to protrude to the maximum extent toward the front of the body **10**, the area of the support polygon of the mobile robot **1** may be increased.

[0160] Therefore, when the front motor **52** rotates, the first ends of the pair of upper arms **56** may move linearly along the front screw **53**. For example, when the front motor **52** rotates, the front screw **53** may rotate. When the front screw **53** rotates, the front nut **54** screw-coupled to the front screw **53** may move linearly along the front screw **53**. Because the first ends of the pair of upper arms **56** are rotatably connected to the front nut **54** by the middle bar **55**, when the front screw **53** rotates, the first ends of the pair of upper arms **56** may move linearly along the front screw **53**.

[0161] When the pair of upper arms **56** move linearly along the front screw **53**, the pair of front drive wheels **20** may be swung at a certain angle with respect to the body **10** by the pair of upper arms **56** and the pair of lower arms **57**. When the front motor **52** rotates in one direction, the pair of front drive wheels **20** may swing upward with respect to the body **10**, that is, counter-clockwise, and rise. When the front motor **52** rotates in the opposite direction, the pair of front drive wheels **20** may swing downward with respect to the body **10**, that is, clockwise, and descend.

[0162] The rear wheel switching device **60** may be configured to move the pair of rear drive wheels **30**. The rear wheel switching device **60** may be configured to move the pair of rear drive wheels **30** back and forth with respect to the body **10**.

[0163] In the four-wheel mode, the rear wheel switching device **60** may move the pair of rear drive wheels **30** toward the rear of the body **10** so that the pair of rear drive wheels **30** protrude from the rear end of the body **10**. The pair of rear drive wheels **30** may be disposed to protrude to the maximum extent toward the rear of the body **10**. When the pair of rear drive wheels **30** are

configured to protrude as much as possible from the rear end of the body **10**, the area of the support polygon of the mobile robot **1** may be increased.

[0164] In the two-wheel mode, the rear wheel switching device **60** may move the pair of rear drive wheels **30** to the center of the body **10** to position the pair of rear drive wheels **30** at the center of the body **10**, thereby lowering the body **10**. Then, the four auxiliary wheels **40** disposed on the lower surface of the body **10** come into contact with the moving surface **100**. At this time, the pair of rear drive wheels **30** may be positioned at the center of the four auxiliary wheels **40**. For example, as illustrated in FIG. **16**, two auxiliary wheels **40** may be positioned on the left side of the pair of rear drive wheels **30**, and the other two auxiliary wheels **40** may be positioned on the right side of the pair of rear drive wheels **30**.

[0165] The rear wheel switching device **60** may include a rear linear actuator **61**. The rear linear actuator **61** may be configured to move the pair of rear drive wheels **30** back and forth with respect to the body **10**. The pair of rear drive wheels **30** may be positioned at the rear of the body **10** or at the center of the body **10** by the rear linear actuator **61**.

[0166] The rear linear actuator **61** may be disposed at the lower portion of the body **10**. For example, the rear linear actuator **61** may be disposed on the upper surface of the lower support plate **12** of the body **10**.

[0167] The body **10** may include an inclined block **15** disposed on the upper surface of the lower support plate **12**. The rear linear actuator **61** may be disposed on an inclined surface **151** of the inclined block **15**.

[0168] The inclined block **15** may be formed to have the inclined surface **151** that is inclined upwardly with respect to the lower support plate **12**. The inclined block **15** may be disposed so that the high point of the inclined surface **151** is adjacent to the front end of the body **10** and the low point of the inclined surface **151** is adjacent to the rear end of the body **10**. The angle of the inclined surface **151** may be defined so that when the pair of rear drive wheels **30** are positioned at the center of the body **10**, the four auxiliary wheels **40** disposed on the lower surface of the body **10** come into contact with the moving surface **100**.

[0169] Referring to FIGS. **5**, **8**, and **10**, the rear linear actuator **61** may include a rear motor **62**, a rear screw **63**, and a rear nut **64**.

[0170] The rear motor **62** may be disposed on the lower support plate **12**. The rear motor **62** may be disposed on a first end of the inclined block **15**. The rear motor **62** may be disposed at the high point of the inclined block **15**. The rear motor **62** may be supported by a motor bracket **623** disposed on the first end of the inclined block **15**.

[0171] The rear motor **62** may be configured to rotate the rear screw **63**. The rear motor **62** may be configured to rotate the rear screw **63** in both directions. The rear motor **62** may include a motor shaft **621**.

[0172] The rear screw **63** may be disposed to be rotated by the rear motor **62**. The rear screw **63** may be disposed so that both ends of the rear screw **63** are supported on the inclined surface **151** of the inclined block **15** disposed on the lower support plate **12**. The rear screw **63** may be supported by a pair of support blocks **631** disposed on the inclined surface **151** of the inclined block **15**.

[0173] The first end of the rear screw **63** may be coupled to the motor shaft **621** of the rear motor **62** by a coupling **622**. Therefore, when the motor shaft **621** rotates, the rear screw **63** may rotate integrally with the motor shaft **621**.

[0174] The rear nut **64** may be screw-coupled to the rear screw **63**. Therefore, when the rear screw **63** rotates, the rear nut **64** may move linearly along the rear screw **63**. For example, the rear screw **63** and the rear nut **64** may be configured to convert the rotational motion of the rear motor **62** into the linear motion of the rear nut **64**. The rear screw **63** and the rear nut **64** may be formed as a lead screw and a nut or a ball screw and a ball nut.

[0175] The linear motion of the rear nut **64** may be guided by an LM guide **65**. The LM guide **65** may be disposed on the upper surface of the inclined block **15**. The LM guide **65** may be disposed

between the pair of support blocks **631**.

[0176] The LM guide **65** may include an LM block **651**. The LM block **651** may be disposed to slide relative to the LM guide **65**. The rear nut **64** may be disposed on the upper surface of the LM block **651**. Therefore, the rear nut **64** may slide linearly along the LM guide **65**.

[0177] The rear wheel switching device **60** may include a support bar **66** configured to connect the pair of rear drive wheels **30** and the rear linear actuator **61**. A first end of the support bar **66** may be connected to the rear linear actuator **61**, and a second end thereof may be connected to the pair of rear drive wheels **30**.

[0178] The support bar **66** may be connected to the rear nut **64**. For example, the rear nut **64** may be disposed at the center of the support bar **66**. Therefore, when the rear nut **64** moves linearly by the rear screw **63**, the support bar **66** may move linearly along the rear screw **63** together with the rear nut **64**.

[0179] The support bar **66** may include a left support bar **661** and a right support bar **662**.

[0180] The first end of the left support bar **661** may be fixed to the rear nut **64**, and the second end thereof may be connected to a support bracket **33** that supports the rear in-wheel motor **31** of the left rear drive wheel **30**.

[0181] The first end of the right support bar **662** may be fixed to the rear nut **64**, and the second end thereof may be connected to a support bracket **33** that supports the rear in-wheel motor **31** of the right rear drive wheel **30**. The right support bar **662** may be disposed symmetrically with respect to the left support bar **661** based on the rear nut **64**.

[0182] Because the left support bar **661** and the right support bar **662** are configured to have the same structure, the left support bar **661** will be described in detail below with reference to FIGS. **10**, **11A**, **11B**, and **11C**.

[0183] FIG. **11A** is a partial view illustrating a left support bar **661** of a rear wheel switching device **60** of a mobile robot **1** according to one or more embodiments of the disclosure. FIG. **11B** is a partial view illustrating a left support bar **661** of a rear wheel switching device **60** of a mobile robot **1** according to one or more embodiments of the disclosure when a rear drive wheel **30** is raised. FIG. **11C** is a partial view illustrating a left support bar **661** of a rear wheel switching device **60** of a mobile robot **1** according to one or more embodiments of the disclosure when a rear drive wheel **30** is lowered.

[0184] Referring to FIGS. **10**, **11A**, **11B**, and **11C**, the left support bar **661** may include a fixed plate **663**, a movable plate **664**, an upper link **665**, a lower link **666**, and a coil spring **667**.

[0185] The fixed plate **663** may be disposed on the rear linear actuator **61**. For example, the fixed plate **663** may be fixed to the rear nut **64** of the rear linear actuator **61**. The fixed plate **663** may be formed in a rectangular flat plate.

[0186] The movable plate **664** may be disposed at a certain distance from the fixed plate **663** and parallel to the fixed plate **663**. The movable plate **664** may be formed in a rectangular flat plate. The rear drive wheel **30** may be disposed on the movable plate **664**. The rear drive wheel **30** may be disposed on the lower portion of the movable plate **664**. The movable plate **664** may be disposed on the support bracket **33** of the rear in-wheel motor **31**.

[0187] The upper link **665** and the lower link **666** may be disposed between the fixed plate **663** and the movable plate **664**.

[0188] A first end of the upper link **665** may be rotatably disposed on the fixed plate **663**, and a second end thereof may be rotatably disposed on the movable plate **664**. The upper link **665** may be formed in a straight bar shape.

[0189] The lower link **666** may be disposed below the upper link **665**. The first end of the lower link **666** may be rotatably disposed on the fixed plate **663**, and the second end thereof may be rotatably disposed on the movable plate **664**. The lower link **666** may be formed in a curved bar shape that is convex upward. When the lower link **666** is formed in the curved bar shape, interference with the lower support plate **12** may be prevented.

[0190] Therefore, the rear drive wheel **30** may move vertically with respect to the rear linear actuator **61** by the upper link **665**, the lower link **666**, and the movable plate **664**.

[0191] The coil spring **667** may be configured to apply a force downward to the rear drive wheel **30**. When the coil spring **667** applies a force downward to the rear drive wheel **30**, the rear drive wheel **30** may maintain contact with the moving surface **100**. Therefore, when the moving surface **100** is uneven, the rear drive wheel **30** may move while maintaining contact with the moving surface **100**.

[0192] The coil spring **667** may be arranged diagonally between the fixed plate **663** and the movable plate **664**. The coil spring **667** may be arranged diagonally between the upper link **665** and the lower link **666**. For example, a first end of the coil spring **667** may be connected to the second end of the upper link **665** disposed on the movable plate **664**, and a second end of the coil spring **667** may be connected to the first end of the lower link **666** disposed on the fixed plate **663**. In this case, a tension spring may be used as the coil spring **667**.

[0193] As another example, the coil spring **667** may further include a shock absorber. The coil spring **667** and the shock absorber may be arranged in a straight line.

[0194] FIG. **11A** illustrates a case where the rear drive wheel **30** moves on a flat surface. In this scenario, the first end and the second end of the lower link **666** may be located on a horizontal plane H. The first end and the second end of the upper link **665** may be located on a plane parallel to the horizontal plane H.

[0195] When the rear drive wheel **30** is located on a convex surface, the rear drive wheel **30** rises as shown in FIG. **11B**.

[0196] As illustrated in FIG. **11B**, when the rear drive wheel **30** rises, the second end of the lower link **666** may rise by a certain height H1 higher than the first end thereof. The second end of the upper link **665** may also rise by the certain height H1 higher than the first end thereof. FIG. **11B** illustrates a state where the rear drive wheel **30** rises about 10 mm. In this case, because the tension coil spring **667** is disposed between the second end of the upper link **665** and the first end of the lower link **666**, the rear drive wheel **30** may receive a downward force by the coil spring **667**. Therefore, the rear drive wheel **30** may maintain contact with the convex surface.

[0197] When the rear drive wheel **30** is located on a concave surface, the rear drive wheel **30** may be descend as illustrated in FIG. **11C**.

[0198] As illustrated in FIG. **11C**, when the rear drive wheel **30** is lowered, the second end of the lower link **666** may be lowered by a certain height H2 lower than the first end thereof. The second end of the upper link **665** may be also lowered by the certain height H2 lower than the first end thereof. FIG. **11C** shows a state where the rear drive wheel **30** is lowered by about 20 mm. In this case, because the tension coil spring **667** is disposed between the second end of the upper link **665** and the first end of the lower link **666**, the rear drive wheel **30** may receive a downward force by the coil spring **667**. Therefore, the rear drive wheel **30** may maintain contact with the concave surface.

[0199] In the above, the first end of the coil spring **667** is connected to the second end of the upper link **665** disposed on the movable plate **664**, and the second end of the coil spring **667** is connected to the first end of the lower link **666** disposed on the fixed plate **663**. However, the arrangement of the coil spring **667** is not limited thereto.

[0200] As another example, as illustrated in FIG. **12**, the coil spring **667** may be arranged in the opposite direction.

[0201] FIG. **12** is a partial view illustrating a left support bar **661** of a rear wheel switching device **60** of a mobile robot **1** according to one or more embodiments of the disclosure.

[0202] Referring to FIG. **12**, the coil spring **667** may be arranged diagonally between the fixed plate **663** and the movable plate **664**. The coil spring **667** may be arranged diagonally between the upper link **665** and the lower link **666**. For example, the first end of the coil spring **667** may be connected to the second end of the lower link **666** disposed on the movable plate **664**, and the

second end of the coil spring **667** may be connected to the first end of the upper link **665** disposed on the fixed plate **663**. In this case, a compression spring may be used as the coil spring **667**.

[0203] Then, the coil spring **667** may apply a force in a downward direction to the rear drive wheel **30**. When the coil spring **667** applies a force in a downward direction to the rear drive wheel **30**, the rear drive wheel **30** may maintain contact with the moving surface **100**.

[0204] The left support bar **661** and the right support bar **662** may be configured as a double wishbone suspension.

[0205] When the rear motor **62** rotates, the support bar **66** may move linearly along the rear screw **63**. For example, when the rear motor **62** rotates, the rear screw **63** may rotate. When the rear screw **63** rotates, the rear nut **64** screw-coupled to the rear screw **63** may move linearly along the rear screw **63**. Because the support bar **66** is connected to the rear nut **64**, when the rear screw **63** rotates, the support bar **66** may move linearly along the rear screw **63**.

[0206] When the support bar **66** moves linearly along the rear screw **63** disposed on the inclined block **15**, the body **10** may be raised or lowered with respect to the moving surface **100** by the inclined block **15**. When the rear motor **62** rotates in one direction, the rear drive wheels **30** may move to the center of the body **10** so that the body **10** may be lowered. When the rear motor **62** rotates in the opposite direction, the rear drive wheels **30** may move rearward so that the body **10** may be raised.

[0207] In the four-wheel mode illustrated in FIGS. **4** to **8**, the pair of front drive wheels **20** and the pair of rear drive wheels **30** come into contact with the moving surface **100**. At this time, the pair of front drive wheels **20** may protrude to the maximum from the front end of the body **10**, and the pair of rear drive wheels **30** may be protrude to the maximum from the rear end of the body **10**. Accordingly, the area of the support polygon of the mobile robot **1** may be increased, so that the mobile robot **1** may stably move on an uneven moving surface **100**, such as outdoors.

[0208] In the four-wheel mode, the four auxiliary wheels **40** may not come into contact with the moving surface **100**. For example, the four auxiliary wheels **40** may be spaced apart from the moving surface **100** by a certain distance.

[0209] The four auxiliary wheels **40** may be disposed on the lower surface of the body **10**. For example, the four auxiliary wheels **40** may be disposed on the lower surface of the lower support plate **12**. The two auxiliary wheels **40** may be disposed adjacent to the front end of the lower support plate **12**. The other two auxiliary wheels **40** may be disposed adjacent to the rear end of the lower support plate **12**.

[0210] The auxiliary wheels **40** may be configured to enable omnidirectional movement so that the mobile robot **1** may rotate in place in the two-wheel mode. In other words, a wheel capable of omnidirectional rotation may be used as the auxiliary wheels **40**. For example, a caster, an omni wheel, a mecanum wheel, etc. may be used as the auxiliary wheels **40**.

[0211] In the case of the embodiment of the disclosure, an omni wheel may be used as the auxiliary wheel **40** as illustrated in FIGS. **13** and **14**.

[0212] FIG. **13** is a perspective view illustrating an auxiliary wheel **40** of a mobile robot **1** according to one or more embodiments of the disclosure. FIG. **14** is a side view illustrating the auxiliary wheel **40** of FIG. **13** according to one or more embodiments of the disclosure.

[0213] Referring to FIGS. **13** and **14**, the auxiliary wheel **40** may include an auxiliary arm **41** and an elastic member **42**. The auxiliary arm **41** may be formed in an approximately L shape. The auxiliary wheel **40** may be rotatably disposed at a first end of the auxiliary arm **41**. The elastic member **42** may be disposed at the second end of the auxiliary arm **41**.

[0214] For example, the auxiliary arm **41** may include a pair of auxiliary arms **41** arranged in parallel. The auxiliary wheel **40** may be rotatably disposed between first ends of the pair of auxiliary arms **41**. The pair of auxiliary arms **41** may be elastically supported by the elastic member **42** disposed at the second ends thereof.

[0215] The auxiliary arm **41** may be rotatably disposed on the lower surface of the lower support

plate **12**. The auxiliary arm **41** may be rotatably supported by an auxiliary shaft **43**. The auxiliary shaft **43** may be supported at both ends by a pair of auxiliary brackets **44**. The pair of auxiliary brackets **44** may be disposed on the lower surface of the lower support plate **12**. Therefore, the auxiliary arm **41** may be rotatably disposed on the lower support plate **12** by the auxiliary shaft **43** and the pair of auxiliary brackets **44**.

[0216] The elastic member **42** may be disposed between the second end of the auxiliary arm **41** and a fixed bracket **45**. For example, a first end of the elastic member **42** may be supported by the second end of the auxiliary arm **41**, and a second end of the elastic member **42** may be supported by the fixed bracket **45**. Accordingly, the auxiliary wheel **40** may be elastically supported with respect to the lower support plate **12** by the elastic member **42**. A coil spring may be used as the elastic member **42**.

[0217] The mobile robot **1** according to one or more embodiments of the disclosure may be converted from the four-wheel mode to the two-wheel mode by the wheel switching devices **50** and **60**. The mobile robot **1** converted to the two-wheel mode is illustrated in FIGS. **15** to **17**.

[0218] FIG. **15** is a perspective view illustrating a mobile robot **1** according to one or more embodiments of the disclosure, which has been switched to a two-wheel mode. FIG. **16** is a side view illustrating the mobile robot **1** of FIG. **15** according to one or more embodiments of the disclosure. FIG. **17** is a front view illustrating the mobile robot **1** of FIG. **15** according to one or more embodiments of the disclosure.

[0219] As illustrated in FIGS. **15** to **17**, when the mobile robot **1** according to one or more embodiments of the disclosure is in the two-wheel mode, the pair of front drive wheels **20** may be spaced apart from the moving surface **100**, and the pair of rear drive wheels **30** and the four auxiliary wheels **40** may be in contact with the moving surface **100**. At this time, the pair of front drive wheels **20** may be raised and positioned inside the body **10**, and the pair of rear drive wheels **30** may be positioned at the center of the body **10**.

[0220] For example, when the front motor **52** rotates, the front screw **53** may rotate. When the front screw **53** rotates, the front nut **54** may move linearly along the front screw **53** toward the center of the body **10**. Because the middle bar **55** to which the pair of upper arms **56** are connected is fixed to the front nut **54**, the pair of upper arms **56** may move linearly along the front screw **53** toward the center of the body **10**.

[0221] When the pair of upper arms **56** move linearly toward the center of the body **10** along the front screw **53**, the front drive wheels **20** may be rotated in a counter-clockwise direction by a certain angle with respect to the body **10** by the pair of upper arms **56** and the pair of lower arms **57**. Then, as illustrated in FIG. **16**, the pair of front drive wheels **20** are spaced apart from the moving surface **100** and positioned between the upper support plate **11** and the lower support plate **12** of the body **10**, and do not protrude from the front surface of the body **10**.

[0222] In addition, when the rear motor **62** rotates, the rear screw **63** may rotate so that the rear nut **64** moves along the rear screw **63** to the center of the body **10**. At this time, the movement of the rear nut **64** may be guided by the LM guide **65** and the LM block **651**. Because the support bar **66** supporting the pair of rear drive wheels **30** is fixed to the rear nut **64**, the pair of rear drive wheels **30** may move to the center of the body **10**.

[0223] Because the LM guide **65** is disposed on the inclined surface **151** of the inclined block **15** disposed on the lower support plate **12** of the body **10**, the rear nut **64** may move along the inclined surface **151**, that is, in a diagonal direction, toward the center of the body **10**. Because the rear nut **64** is provided with the support bar **66** that supports the pair of rear drive wheels **30**, the pair of rear drive wheels **30** may move along the inclined surface **151**, that is, in the diagonal direction.

[0224] When the pair of rear drive wheels **30** move along the inclined surface **151** of the inclined block **15** of the body **10** to the center of the body **10**, the pair of rear drive wheels **30** may remain in contact with the moving surface **100** and the body **10** may move downward. When the pair of rear drive wheels **30** are positioned at the center of the body **10**, the four auxiliary wheels **40** disposed

on the lower surface of the body **10** may descend and come into contact with the moving surface **100**. For example, in the two-wheel mode, the pair of rear drive wheels **30** and the four auxiliary wheels **40** may come into contact with the moving surface **100**. At this time, two auxiliary wheels **40** may be positioned in front of the pair of rear drive wheels **30**, and the other two auxiliary wheels **40** may be positioned in the rear of the pair of rear drive wheels **30**.

[0225] When the mobile robot **1** is in the two-wheel mode, the area of the support polygon of the mobile robot **1** may be minimized, so that the mobile robot **1** may efficiently move in a narrow or complicated place such as indoors. In particular, in the two-wheel mode, the mobile robot **1** may rotate in place by the pair of rear drive wheels **30**.

[0226] FIG. **18** is a block diagram illustrating a mobile robot **1** according to one or more embodiments of the disclosure.

[0227] Referring to FIG. **18**, the mobile robot **1** according to one or more embodiments of the disclosure may include a processor **90**.

[0228] The processor **90** may be configured to control the mobile robot **1** in whole or in part. For example, the processor **90** may be configured to control the wheel switching devices **50** and **60** of the mobile robot **1**. As an example, the processor **90** may control the wheel switching devices **50** and **60** to switch the mobile robot **1** to the four-wheel mode or the two-wheel mode. For example, when the mobile robot **1** is in the four-wheel mode, the processor **90** may control the wheel switching devices **50** and **60** to switch the mobile robot **1** to the two-wheel mode. When the mobile robot **1** is in the two-wheel mode, the processor **90** may control the wheel switching devices **50** and **60** to switch the mobile robot **1** to the four-wheel mode.

[0229] The processor **90** may be configured in various ways as long as it can control the mobile robot **1**. For example, the processor **90** may be implemented as a microprocessor, a graphics-processing unit (GPU), an artificial intelligence (AI) processor, a neural processing unit (NPU), or a time controller (TCN). However, the processor **90** is not limited thereto, and may include one or more of a central processing unit (CPU), a micro controller unit (MCU), a micro processing unit (MPU), a controller, an application processor (AP), a communication processor (CP), and an advanced reduced instruction set computing (RISC) machine (ARM) processor or may be defined as a corresponding term. In addition, the processor **90** may be implemented as a system on chip (SoC) or a large scale integration (LSI) in which a processing algorithm is built therein, or may be implemented as an application specific integrated circuit (ASIC) type or a field programmable gate array (FPGA) type.

[0230] The mobile robot **1** may include a memory **91**. The memory **91** may store map data. The map data may include information about the outdoors and information about the indoors. The map data may include information about the moving surface **100**. For example, the map data may include a reference height of a step on which the mobile robot **1** travels in the four-wheel mode.

[0231] In addition, the memory **91** may store at least one instruction regarding the mobile robot **1**. The memory **91** may store an operating system (O/S) for driving the mobile robot **1**. In addition, the memory **91** may store various software programs or applications for operating the mobile robot **1** according to one or more embodiments of the disclosure.

[0232] The memory **91** may include a semiconductor memory such as a flash memory and the like or a magnetic storage medium such as a hard disk and the like.

[0233] On the other hand, in the disclosure, the term memory **91** may be used to refer to a memory, a read-only memory (ROM) within the processor, a random-access memory (RAM), or a memory card (e.g., a micro SD card, a memory stick) mounted on the mobile robot **1**.

[0234] The processor **90** may include one or multiple processors. For example, the processor **90** may perform an operation of the mobile robot **1** according to one or more embodiments of the disclosure by executing at least one instruction stored in the memory **91**.

[0235] The mobile robot **1** may include a user interface **92**. The mobile robot **1** according to the embodiment of the disclosure may use a touch display as the user interface **92**. The mobile robot **1**

may output the status of the mobile robot **1** through the touch display. In addition, the mobile robot **1** may receive user instructions through the touch display. For example, the mobile robot **1** may receive a four-wheel mode command or a two-wheel mode command input by the user through the touch display.

[0236] However, the user interface **92** used in the mobile robot **1** of the disclosure is not limited thereto. The user interface **92** may include various types of input devices. For example, the user interface **92** may include physical buttons. In this case, the physical buttons may include a function key, a direction key, and a dial button. The physical button may be implemented with the plurality of keys.

[0237] The mobile robot **1** may include a communication interface **93**. The mobile robot **1** may communicate with an external device, such as a server, a mobile device, and the like, through the communication interface **93**. The mobile robot **1** may receive an operation mode command from an external device through the communication interface **93**.

[0238] The processor **90** may control the mobile robot **1** according to an instruction received through the communication interface **93**.

[0239] The communication interface **93** may be implemented as wireless communication. For example, the communication interface **93** may be configured to perform communication using at least one of the following communication methods: Wi-Fi, Wi-Fi Direct, Bluetooth, Zigbee, third generation (3G), third generation partnership project (3GPP), long term evolution (LTE), and fifth generation (5G).

[0240] The mobile robot **1** may include a pair of front in-wheel motors **21** and a pair of rear in-wheel motors **31**. The front in-wheel motor **21** may be disposed in the front drive wheel **20** to drive the front drive wheel **20**. The rear in-wheel motor **31** may be disposed in the rear drive wheel **30** to drive the rear drive wheel **30**. The processor **90** may control the pair of front in-wheel motors **21** and the pair of rear in-wheel motors **31** to move the mobile robot **1** in a desired direction.

[0241] The mobile robot **1** may include a traveling sensor **95**. The processor **90** may use a signal input from the traveling sensor **95** to move the mobile robot **1** to a destination. The processor **90** may use the traveling sensor **95** to avoid obstacles on the moving path. The processor **90** may use the traveling sensor **95** to recognize the state of the moving path. For example, the processor **90** may use the traveling sensor **95** to recognize whether the mobile robot **1** is indoors or outdoors.

[0242] The traveling sensor **95** may include an ultrasonic sensor, a three dimensional camera (3D camera), a time of flight sensor (TOF sensor), a light detection and ranging sensor (LiDAR sensor), etc. The processor **90** may recognize the position of the mobile robot **1** using a signal input from the traveling sensor **95**. The processor **90** may recognize an obstacle on the moving path using a signal input from the traveling sensor **95**.

[0243] The mobile robot **1** may include a microphone **96**. The processor **90** may control the mobile robot **1** using a signal input from the microphone **96**. The microphone **96** may be configured to convert a user's voice into an electric signal. Therefore, the user may control the mobile robot **1** by voice.

[0244] The mobile robot **1** may include a speaker **97**. The processor **90** may output an alarm or information about the status of the mobile robot **1** as a sound through the speaker **97**.

[0245] The mobile robot **1** may include a power supply **98**. The power supply **98** may be configured to supply power to all internal components of the mobile robot **1**. For example, a rechargeable battery may be used as the power supply **98**.

[0246] The processor **90** may recognize the status of the power supply **98** using a signal input from the power supply **98**. The processor **90** may output the status of the power supply **98** through the speaker **97**. When the power of the power supply **98** is insufficient, the processor **90** may control the pair of front in-wheel motors **21** and the pair of rear in-wheel motors **31** to move the mobile robot **1** to a charging station.

[0247] When the processor **90** recognizes that the mobile robot **1** is located outdoors using the

traveling sensor **95**, the processor **90** may cause the mobile robot **1** to move in the four-wheel mode. When the current mode of the mobile robot **1** is the two-wheel mode, the processor **90** may control the front motor **52** of the front wheel switching device **50** and the rear motor **62** of the rear wheel switching device **60** to switch the two-wheel mode to the four-wheel mode. After that, the processor **90** may control the pair of front in-wheel motors **21** and the pair of rear in-wheel motors **31** to travel the mobile robot **1**.

[0248] When the processor **90** recognizes that a space where the mobile robot **1** is located is narrow or complex based on the signal from the traveling sensor **95**, the processor **90** may cause the mobile robot **1** to travel in the two-wheel mode. When the current mode of the mobile robot **1** is the four-wheel mode, the processor **90** may control the front motor **52** of the front wheel switching device **50** and the rear motor **62** of the rear wheel switching device **60** to switch the four-wheel mode to the two-wheel mode. After that, the processor **90** may control the pair of rear in-wheel motors **31** to travel the mobile robot **1**.

[0249] When the processor **90** recognizes that the mobile robot **1** has passed through the narrow or complex space using the traveling sensor **95**, the processor **90** may cause the mobile robot **1** to travel in the four-wheel mode. For example, the processor **90** may control the wheel switching devices **50** and **60** to switch the two-wheel mode to the four-wheel mode, and may control the pair of front in-wheel motors **21** and the pair of rear in-wheel motors **31** to travel the mobile robot **1**.

[0250] When the processor **90** recognizes that the mobile robot **1** is located indoors using the traveling sensor **95**, the processor **90** may cause the mobile robot **1** to travel in the two-wheel mode. When the current mode of the mobile robot **1** is the four-wheel mode, the processor **90** may control the wheel switching devices **50** and **60** to switch the four-wheel mode to the two-wheel mode, and may control the pair of rear in-wheel motors **31** to travel the mobile robot **1**.

[0251] During traveling, when the processor **90** recognizes a high step on the moving surface **100** from a signal of the traveling sensor **95**, the processor **90** may control the wheel switching devices **50** and **60** to switch the two-wheel mode to the four-wheel mode. After the mobile robot **1** passes the high step in the four-wheel mode, the processor **90** may control the wheel switching devices **50** and **60** to switch the four-wheel mode to the two-wheel mode to travel the mobile robot **1**.

[0252] When the processor **90** recognizes that excessive torque has occurred in the front in-wheel motor **21** and/or the rear in-wheel motor **31**, the processor **90** may control the front motor **52** of the front wheel switching device **50** and the rear motor **62** of the rear wheel switching device **60** to switch the two-wheel mode to the four-wheel mode. When the mobile robot **1** passes over an obstacle such as a bump, a carpet, etc., excessive torque may occur in the front in-wheel motor **21** and/or the rear in-wheel motor **31**. After the mobile robot **1** passes over the obstacle in the four-wheel mode, the processor **90** may control the wheel switching devices **50** and **60** to switch the four-wheel mode to the two-wheel mode to travel the mobile robot **1**.

[0253] The mobile robot **1** according to one or more embodiments of the disclosure having the structure as described above may travel in the four-wheel mode capable of exerting a strong driving force outdoors, and may travel in the two-wheel mode capable of exerting a fast steering performance indoors. Therefore, the mobile robot **1** according to one or more embodiments of the disclosure may achieve high traveling performance in various indoor and outdoor environments.

[0254] In addition, the mobile robot **1** according to one or more embodiments of the disclosure may travel in the four-wheel mode when encountering an obstacle while traveling indoors, thereby improving obstacle overcoming performance.

[0255] In addition, the mobile robot **1** according to one or more embodiments of the disclosure may rotate in place in the two-wheel mode while traveling indoors, thereby effectively passing through narrow or complex spaces.

[0256] In the foregoing, the disclosure has been shown and described with reference to example embodiments. However, it is understood by those skilled in the art that various changes may be

made in form and detail without departing from the scope of the disclosure as defined by the appended claims and equivalents thereof.

Claims

- 1.** A mobile robot comprising: a body; a pair of front drive wheels at a front portion of the body; a pair of rear drive wheels at a rear portion of the body; four auxiliary wheels on a lower surface of the body; a front wheel switching device comprising a front linear actuator disposed at an upper portion of the body; and a rear wheel switching device comprising a rear linear actuator disposed at a lower portion of the body, wherein the front wheel switching device and the rear wheel switching device are configured to switch the pair of front drive wheels and the pair of rear drive wheels between a four-wheel mode and a two-wheel mode, wherein, in the four-wheel mode, the pair of front drive wheels and the pair of rear drive wheels are in contact with a moving surface, and the four auxiliary wheels are spaced apart from the moving surface, and wherein, in the two-wheel mode, the pair of front drive wheels are spaced apart from the moving surface, the pair of rear drive wheels are in a center of the body, and the pair of rear drive wheels and the four auxiliary wheels are in contact with the moving surface.
- 2.** The mobile robot of claim 1, wherein the front wheel switching device is configured to move the pair of front drive wheels up and down with respect to the body; and wherein the rear wheel switching device is configured to move the pair of rear drive wheels forward and backward with respect to the body.
- 3.** The mobile robot of claim 2, wherein the front linear actuator is configured to move the pair of front drive wheels up and down.
- 4.** The mobile robot of claim 3, wherein the front wheel switching device further comprises: an upper arm comprising a first end connected to the front linear actuator and a second end connected to the pair of front drive wheels; and a lower arm comprising a first end connected to the lower portion of the body and a second end connected to the pair of front drive wheels.
- 5.** The mobile robot of claim 4, wherein the front linear actuator comprises: a front motor; a front screw configured to be rotated by the front motor; and a front nut that is screw-coupled to the front screw and to which the first end of the upper arm is connected, and wherein rotation of the front motor causes the first end of the upper arm to move linearly along the front screw.
- 6.** The mobile robot of claim 4, wherein the upper arm comprises a suspension.
- 7.** The mobile robot of claim 6, wherein the suspension comprises a coil spring and a shock absorber.
- 8.** The mobile robot of claim 2, wherein the rear linear actuator is configured to move the pair of rear drive wheels forward and backward with respect to the body.
- 9.** The mobile robot of claim 8, wherein the body comprises: an upper support plate; a lower support plate below the upper support plate; and an inclined block on an upper surface of the lower support plate, the inclined block including an inclined surface, and wherein the rear linear actuator is on the inclined surface of the inclined block.
- 10.** The mobile robot of claim 8, wherein the rear wheel switching device further comprises a support bar comprising a first end connected to the rear linear actuator and a second end connected to the pair of rear drive wheels.
- 11.** The mobile robot of claim 10, wherein the rear linear actuator comprises: a rear motor; a rear screw configured to be rotated by the rear motor; and a rear nut which is screw-coupled to the rear screw and is connected to the support bar, and wherein rotation of the rear motor causes the support bar to move linearly along the rear screw.
- 12.** The mobile robot of claim 8, wherein the rear wheel switching device further comprises: a fixed plate on the rear linear actuator; a movable plate parallel to the fixed plate and spaced apart from the fixed plate by a certain distance, wherein the pair of rear drive wheels are on the moveable

plate; an upper link comprising a first end rotatably attached to the fixed plate and a second end rotatably attached to the movable plate; a lower link below the upper link, the lower link comprising a first end rotatably attached to the fixed plate and a second end rotatably attached to the movable plate; and a coil spring between the upper link and the lower link.

13. The mobile robot of claim 8, wherein the rear wheel switching device comprises: a fixed plate on the rear linear actuator; a movable plate parallel to the fixed plate and spaced apart from the fixed plate by a certain distance, wherein one of the pair of rear drive wheels is on the moveable plate; and a suspension between the movable plate and the fixed plate.

14. The mobile robot of claim 1, wherein each of the pair of front drive wheels and each of the pair of rear drive wheels comprises an in-wheel motor.

15. The mobile robot of claim 1, wherein each of the four auxiliary wheels comprises an omni wheel.

16. A mobile robot comprising: a body; a pair of front drive wheels at a front portion of the body; a pair of rear drive wheels at a rear portion of the body; a plurality of auxiliary wheels on a lower surface of the body; a front wheel switching device comprising: a front motor; a front screw configured to be rotated by the front motor; a front nut that is screw-coupled to the front screw; an upper arm comprising a first end connected to the front nut and a second end connected to the pair of front drive wheels; and a lower arm comprising a first end connected to a lower portion of the body and a second end connected to the pair of front drive wheels; and a rear wheel switching device comprising: a rear motor; a rear screw configured to be rotated by the rear motor; a rear nut which is screw-coupled to the rear screw; and a support bar connected to the rear nut and the pair of rear drive wheels, wherein rotation of the front motor causes the first end of the upper arm to move linearly along the front screw, and wherein rotation of the rear motor causes the support bar to move linearly along the rear screw.

17. The mobile robot of claim 16, wherein linear movement of the first end of the upper arm along the front screw causes the pair of front drive wheels to move up or down with respect to the body, and wherein linear movement of the support bar along the rear screw causes the pair of rear drive wheels to move up and forward, or down and backward, with respect to the body.

18. The mobile robot of claim 16, wherein the upper arm comprises a coil spring and a shock absorber.

19. The mobile robot of claim 16, wherein the body comprises: an upper support plate; a lower support plate below the upper support plate; and an inclined block on an upper surface of the lower support plate, the inclined block including an inclined surface, and wherein the rear screw is on the inclined surface of the inclined block.

20. A mobile robot comprising: a body; a pair of front drive wheels at a front portion of the body; a pair of rear drive wheels at a rear portion of the body; a plurality of auxiliary wheels on a lower surface of the body; a front wheel switching device comprising: a front motor; a front screw configured to be rotated by the front motor; a front nut that is screw-coupled to the front screw; an upper arm comprising a first end connected to the front nut and a second end connected to the pair of front drive wheels; and a lower arm comprising a first end connected to a lower portion of the body and a second end connected to the pair of front drive wheels; and a rear wheel switching device comprising: a rear motor; a rear screw configured to be rotated by the rear motor; a rear nut which is screw-coupled to the rear screw; and a support bar connected to the rear nut and the pair of rear drive wheels, memory storing one or more instructions; and at least one processor configured to execute the one or more instructions, wherein the one or more instructions, when executed by the at least one processor, cause the mobile robot to: cause the front nut to move linearly along the front screw by controlling the front motor to rotate the front screw, wherein linear movement of the front nut causes the pair of front drive wheels to move up or down with respect to the body, and cause the support bar to move linearly along the rear screw by controlling the rear motor to rotate the rear screw, wherein linear movement of the rear nut along the rear

screw causes the pair of rear drive wheels to up and forward, or down and backward, with respect to the body, and wherein, when the pair of front drive wheels move upward with respect to the body, and the pair of rear drive wheels move up and forward with respect to the body, the plurality of auxiliary wheels make contact with a surface in contact with the pair of rear drive wheels.
